

# Product Manual

Kamoer Fluid Tech (Shanghai) Co.,Ltd.

Version: A/2

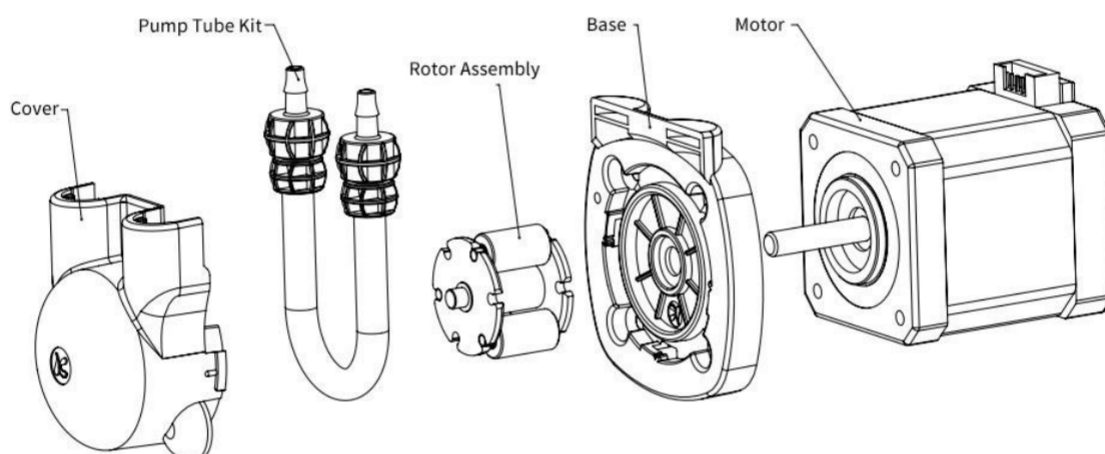
|              |                                       |
|--------------|---------------------------------------|
| Product name | Micro peristaltic pump                |
| Model        | KPAS100                               |
| Release Date | 2022.11.23                            |
| Manufacturer | Kamoer Fluid Tech (Shanghai) Co.,Ltd. |

## A. Product Overview

### 1. Product Images



### 2. Product assembly drawing



### 3. Model Interpretation

|         |            |               |           |                  |                 |                     |
|---------|------------|---------------|-----------|------------------|-----------------|---------------------|
| KPAS100 | ST         | B             | 14        | 3                | A               | A                   |
| Model   | Motor type | Tube material | Tube size | Number of rotors | Connector model | Installation method |

#### 3.1 Motor Type

ST: 24V stepper motor

#### 3.2 Tube type

B: BPT tube      S: Silicone tube



## 3.3 Tube Code

14#: 1.6mm(ID)×4.8mm(OD)    19#: 2.4mm(ID)×5.6mm(OD)    16#: 3.2mm(ID)×6.4mm(OD)

## 3.4 Number of rotors

3: 3 rotors

## 3.5 Connector type

A: Butt joint



B: Clamp(Only supports 14# pump tube)



## 3.6 Installation method

A: Straight board (default installation)    L: L plate

| Model   | Cover | Base | Rotor                | Rotor Assembly | Connector |
|---------|-------|------|----------------------|----------------|-----------|
| KPAS100 | PA    | PA   | Engineering plastics | POM            | PP        |

# 4. Features & Applications

## 4.1 Features

- KPAS100 series peristaltic pump is a stepper motor driven peristaltic pump
- Pump tube selection - the pump tube adopts imported BPT (B) tube and silicone tube (S), which can be applied to different transmission liquid requirements
  - Pump tube size is optional (14#, 19#, 16#) to meet different flow requirements
  - The pump body comes with a 4-hole mounting plate, no need to add it separately
  - 3 rotor transmission, moderate pulse
  - Installation method - The installation method is fixed through the plate. In addition, the pump head adopts a quick-release type, and the orientation of the pipe joint can be selected in any direction of up, down, left, and right
    - Transmission mode - planetary friction reduction transmission
    - Suitable for wall thickness 1.6 pump tubing



## 4.2 Typical applications

Analytical instruments: ion chromatograph, blood analyzer, hypochlorous acid water generator;  
detergent delivery: automatic car washer.



Ion chromatograph



Auto car washer



Blood analyzer



Hypochlorous Acid

- The hose is resistant to liquid media. It is necessary to check chemical compatibility or perform immersion experiments. Improper hose selection may cause the hose to be damaged quickly.
- The working environment of the product should not exceed 45 °C, and the humidity should not exceed 70% (no condensing water). The harsh working environment will cause premature damage to the product.
- Fluid spills due to ruptured hoses depend on the fluid medium and your specific application
- High overload work may lead to premature product failure

## 5. Intellectual property rights, Product certification

CN 307575335S



CE 认证

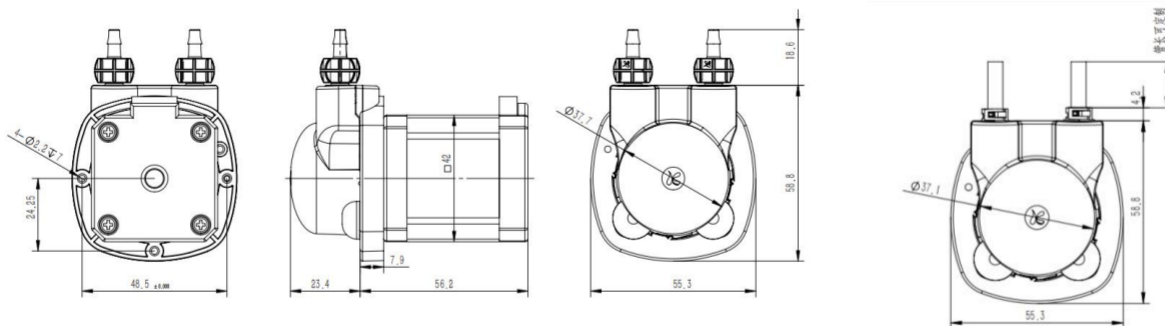


RoHS 认证



## B. Product Specifications

### 1. Dimensions

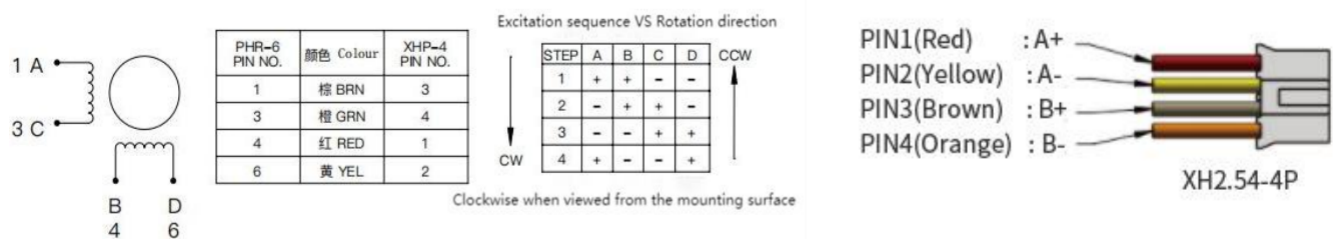


### 2. Parameters

| Item             |                               |          | Technical Parameter                                                                                                                                                                                                                                                                                                                                                                  |                 |                 |                 |                 |                 |
|------------------|-------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Basic parameters | Motor type                    |          | Stepper motor                                                                                                                                                                                                                                                                                                                                                                        |                 |                 |                 |                 |                 |
|                  | Flow range                    |          | 32-110ml/min                                                                                                                                                                                                                                                                                                                                                                         |                 |                 |                 |                 |                 |
|                  | Power                         |          | 20W                                                                                                                                                                                                                                                                                                                                                                                  |                 |                 |                 |                 |                 |
|                  | Control mode                  |          | Stepper driver                                                                                                                                                                                                                                                                                                                                                                       |                 |                 |                 |                 |                 |
|                  | Weight                        |          | About 400g                                                                                                                                                                                                                                                                                                                                                                           |                 |                 |                 |                 |                 |
|                  | Rated voltage                 |          | 24V                                                                                                                                                                                                                                                                                                                                                                                  |                 |                 |                 |                 |                 |
|                  | Pump tube life                |          | S: 200h      BPT: 1000h                                                                                                                                                                                                                                                                                                                                                              |                 |                 |                 |                 |                 |
|                  | Motor life                    |          | 6000h                                                                                                                                                                                                                                                                                                                                                                                |                 |                 |                 |                 |                 |
| Flow Rate        | Reference flow (unit: ml/min) |          |                                                                                                                                                                                                                                                                                                                                                                                      |                 |                 |                 |                 |                 |
|                  | Pump tube code                |          | S14                                                                                                                                                                                                                                                                                                                                                                                  | S19             | S16             | B14             | B19             | B16             |
|                  | ID x OD                       | Metric   | 1.6mm×<br>4.8mm                                                                                                                                                                                                                                                                                                                                                                      | 2.4mm×<br>5.6mm | 3.2mm×<br>6.4mm | 1.6mm×<br>4.8mm | 2.4mm×<br>5.6mm | 3.2mm×<br>6.4mm |
|                  |                               | Imperial | 1/16in×3/16in                                                                                                                                                                                                                                                                                                                                                                        | 3/32in×7/32in   | 1/8in×1/4in     | 1/16in×3/16in   | 3/32in×7/32in   | 1/8in×1/4in     |
|                  | Pump tube material            |          | Silicone tube                                                                                                                                                                                                                                                                                                                                                                        |                 |                 | BPT             |                 |                 |
|                  | Motor speed                   |          | 350 rpm                                                                                                                                                                                                                                                                                                                                                                              |                 |                 |                 |                 |                 |
|                  | ST/Current: 1.2A              |          | 32 ml/min                                                                                                                                                                                                                                                                                                                                                                            | 70 ml/min       | 110 ml/min      | 32 ml/min       | 70 ml/min       | 110 ml/min      |
| Notes            | Remark                        |          | The above parameters are measured with pure water without pressure under the standard atmospheric pressure at room temperature of 20°C. Actually, there will be certain errors in the flow rate according to different media, different outlet pressures, and DC motor speed errors. The data are only for reference. In addition, it can be customized according to customer needs. |                 |                 |                 |                 |                 |
|                  | Reference noise value         |          | In a quiet room with an ambient noise of 35dB, the product is 50cm away from the noise tester, and the noise is 50dB.                                                                                                                                                                                                                                                                |                 |                 |                 |                 |                 |



## 3. Stepper Motor Parameters & Wiring



| Phase | Step angle | Starting current | Insulation class |
|-------|------------|------------------|------------------|
| 2     | 1.8°       | 1.2A             | B                |

## 4. Flow curve



Flow curve is for reference only, and the specific parameters are subject to the actual working condition test.

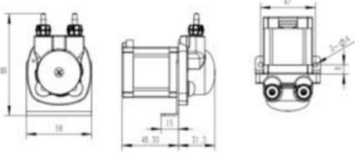




## C. Pump Tube

| Pump tube material | Characteristic                                                                                    | Standards compliant                                                  | Tube code                        |
|--------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------|
| Silicone Tube (S)  | Low adsorption, good temperature resistance, good chemical corrosion resistance, life time >200h  | RoHS<br>Compliant with food grade certification<br>(GB 4806.11-2016) | 01 # 02 # 04# 05#<br>06# 08# 10# |
| BPT tube (B)       | Good resistance to common chemicals, acid and alkali, and oxidation resistance, life time >1000h. | RoHS                                                                 | 04# 12# 06# 08#                  |

## D. Optional accessories

| Accessory name                  | Specifications                 |                                                                                     | Function                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|--------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anti-floating countersunk head  | Diameter 6mm<br>Diameter 2.5mm |    | Place the inlet of the pump tube completely in the liquid to ensure that the peristaltic pump tube can fully absorb the liquid                                                                                                                                                                                           |
| Mounting plate                  | L shape                        |  | Increase the types of installation methods to meet different customer needs                                                                                                                                                                                                                                              |
| Peristaltic pump stepper driver | MODBUS-RTU                     |  | It can realize built-in and external potentiometer speed regulation, can realize MODBUS RS485 communication and 0-10V analog speed regulation, realize electric and trigger two start and stop modes, the maximum subdivision is 32, and the drive current can be set (0.7A, 1.2A) , 1.7A, 2.2A, 2.7A, 2.9A, 3.2A, 4.0A) |

Special declaration: The company reserves the right to improve the appearance and technical specifications of the product without prior notice.

